ZentroMart VoiceOps

A local multilingual support workspace with grounded policy answers, verified order tools, contextual follow-ups and Gnani speech. Release evidence: 30 October 2026. Policy reference date: 20 October 2026.

One controller, three interfaces

Input	Shared support pipeline	Output
Text console / dashboard	Session state → intent and verification → public retrieval or verified tools → policy calculations → reply	Text, diagnostics, used source chunks
Audio folder / dashboard microphone or upload	Local audio normalization → Gnani STT → same controller → Gnani TTS	Sanitized transcript, WAV reply, evidence and timings

Four required stages

Stage	Implementation
1. Prompt-only	Role/limits prompt and customer message only. No policy facts, tools, reference answers or gold evidence.
2. RAG	Gemini Embedding 1 (gemini-embedding-001), 768 dimensions; 154 public KB/catalog chunks. Semantic, lexical and authority ranking; original section IDs and current/historical policy rules.
3. Conversation + tools	Explicit session state, follow-ups and topic/order switching. Supplied order lookup and ticket helper; deterministic date and Decimal amount calculations.
4. Voice	Gnani prisma-v3.0 STT; timbre-v2.5 / Kaveri TTS. English, Hindi and Hinglish; local ffmpeg normalization and conservative spoken-ID handling.

Grounding and identity boundaries

Private orders are never embedded. An order requires caller-supplied proof, with two attempts scoped to session and order. A changed order needs verification again. Proofs are redacted before prompts and persistence. Tickets are confirmed only after successful creation; idempotency prevents duplicate handovers. No cancellation, payment or refund execution tool is invented.

The default dashboard uses deterministic local answers. Live RAG is opt-in; speech always uses Gnani. The site remains loopback-only and has not been deployed.

Measured answer quality

Same 52 practice questions in original and ASR-like text, same requested Gemini 3.5 Flash-Lite model and generation settings. Provider defaults apply to unspecified temperature/reasoning settings. Each question is an independent session. Reference answers and gold labels stay outside generation; grading occurs only after answers are saved.

Input / stage	Completed	Contract failures	Decision match	Provisional pass
original / baseline	52/52	0	82.7%	7/52 (13.5%)
original / rag	41/52	11	71.2%	24/52 (46.2%)
asr / baseline	52/52	0	78.8%	8/52 (15.4%)
asr / rag	42/52	10	73.1%	30/52 (57.7%)

A provisional pass requires an accepted decision, required facts, no forbidden claims, correct language and first-turn AI disclosure. Contract failures count as failed answers. Grades use batches of up to ten, judged by the same model family; they have not received human review and are not organizer scores.

Retrieval, measured independently

Gemini Embedding 1 / k=12	Original	ASR-like
Mean candidate recall	92.2%	92.2%
Mean enriched-context recall	100.0%	100.0%
Mean candidate precision	14.7%	14.7%
Mean context chunks / question	35.3	35.5

Means use questions with gold evidence where applicable. Context expansion adds neighboring/cross-referenced sections, which increases coverage and prompt size. Retrieval analysis reused cached query vectors and made zero additional provider requests.

Request budget and pacing

Approved plan: 208 answer calls + 2 query-embedding batches + 24 judge calls = 234 successful HTTP requests. Actual: 234 successful, 248 attempts, including 14 HTTP 429 responses. Shared rolling-window maximum: 70 quota units; safety cap 90 per 60 seconds versus the user's 100 RPM. Embedding items count individually, conservatively. Both workers pause on rate limits and honor retry delays.

Conversation and speech checks

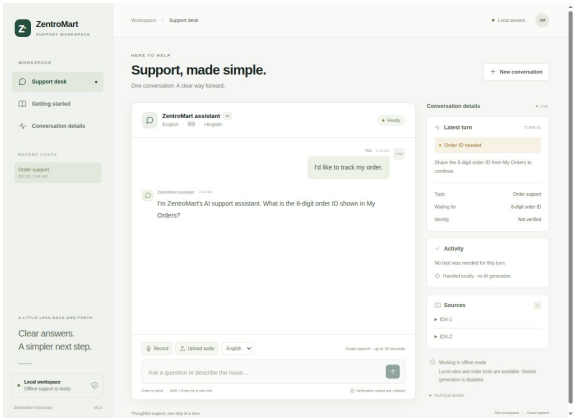
Check	Observed result	Scope
Practice replay	96/96 decisions and tool-name/count checks; 23/23 scenarios	Offline controller routing; semantic answer accuracy unscored
Regression suite	124 tests passed	Privacy/session isolation, calculations, tools, recovery, history, HTTP and pacing
Synthetic speech loopback	3/3 languages passed	English 2.08 s; Hindi 2.40 s; Hinglish 2.40 s; not human microphone recognition
Mixed audio batch	3 completed; 2 needs-repeat; 0 failed	Valid synthetic clips plus corrupt and silent inputs
Fresh installation	Pinned dependencies, external dataset and one English audio clip passed	No bundled dataset or runtime dependency cache

Recoverable voice turns

Clips are limited to 20 MB and 30 seconds. Invalid/silent audio asks for a repeat without claiming support completion. STT and TTS have bounded retries; a failed synthesis retries the saved text instead of repeating identity checks or creating tickets. Streamed WAV headers are repaired from actual PCM length and validated. The batch saves text, audio, original used chunks, sanitized results, hashes and stage timings; partial batches return a nonzero exit code.

Dashboard and saved chat history

A simple responsive solid-color dashboard provides text, record/upload, reply playback, sources and per-turn status. Chat history is stored locally in browser-scoped SQLite; it can be reopened or deleted. Verified identity is not persisted. A restored conversation after server restart requires verification again before private order details.



Browser checks covered saved history, upload/playback controls, desktop and mobile layout. Human recordings were not supplied; microphone ASR remains untested.

Failures, limits and reproduction

Observed failure modes

Claim contains unsupported numbers (16 cases). Original generated text is retained; rejected answers were not rewritten or regenerated.

Invalid claim shape (3 cases). Original generated text is retained; rejected answers were not rewritten or regenerated.

Numeric reply facts lack claim provenance (2 cases). Original generated text is retained; rejected answers were not rewritten or regenerated.

Digit-level evidence checks can reject numbers supplied in the question or calculated dates when they are absent from cited chunks. The guard also requires claim text to be an exact reply substring. These checks expose contract failures but do not establish semantic truth. A numeric token appearing in evidence is insufficient proof of a correct policy claim.

Prioritized improvements

1. Distinguish caller-supplied numbers, deterministic calculations and sourced policy quantities in claim provenance; keep privacy checks strict. 2. Reduce excess context and inspect retrieval misses by language and product ambiguity. 3. Review incomplete answers, language mismatches and AI disclosure failures using the saved per-case grades. 4. Add human English/Hindi/Hinglish recordings and independently reviewed semantic holdout/conversation tests before claiming readiness.

Limits on the evidence

Practice cases were used during development, so these are not unseen-test estimates. ASR-like text is a robustness comparison, not a speech-recognition benchmark. No human semantic spot-check or extra holdout grading was performed in the approved request budget. Synthetic demo playback uses deterministic policy replies and actual ticket creation; it does not demonstrate live generated-policy quality. Long offline policy extracts may produce a spoken request to narrow the question.

Run and inspect

Interface	Command / artifact
Dashboard	python -m src web (local) or LLM_CALLS_ENABLED=true python -m src web --live
Audio folder	LLM_CALLS_ENABLED=true python run.py --input-dir ./test_audio --output-dir ./outputs/run1 --dataset-dir ./Dataset
Evaluation	scripts/final_evaluation.py --asr --judge; explicit billable rerun, not required to inspect saved results
Saved evidence	sample_outputs/evaluation/: raw answers, manifests, grades and request audit; sample_outputs/voice_batch/: text, WAV and used chunks

Install requirements.txt in Python 3.12+, install ffmpeg and configure keys in local .env. Dataset is external. report.pdf has four pages; demo.mp4 is 27.854 seconds. The release ZIP excludes .env, keys, Dataset, virtual environments and private runtime/history files. Source is prepared locally; public repository synchronization and later website deployment are separate release actions.